LEARNER GUIDE

Area of rectangles and triangles

Perimeter and Area · Lesson 2. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A rectangle is rows of equal squares, so its **area = length × width**. A triangle sits inside a rectangle with the same **base** and the same **height** (the straight-up distance from the base line to the top corner, at right angles to the base). Cut the triangle along its height and turn the two pieces: they fill the rest of the rectangle exactly. So every triangle is half its rectangle: **area = ½ × base × height**.

What you are learning to notice

- Rectangle: length × width. Triangle: ½ × base × height.
- The height is perpendicular to the base. It is not a slanted side.
- For an obtuse triangle, the height is measured outside it, but the rule is the same.
- Triangles with the same base and height have the same area, however different they look.
- Areas can be halves: ½ × 5 × 3 = 7½ m^2.

Moving through it, one step at a time

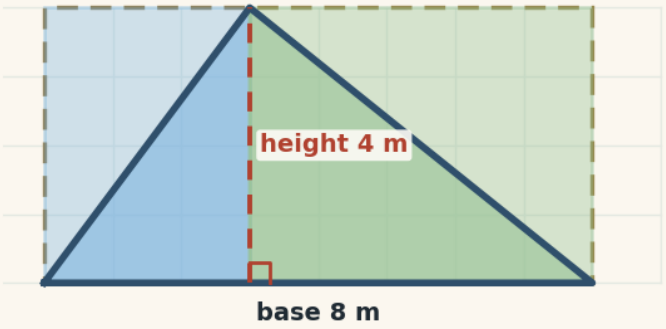
For a rectangle, multiply the length by the width. For a triangle, find the base and the perpendicular height (at right angles to the base), multiply them to get the rectangle round the triangle, then halve. Give the answer in square units (m^2).

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Cut and turn

Find the area of a triangle with base 8 m and height 4 m.

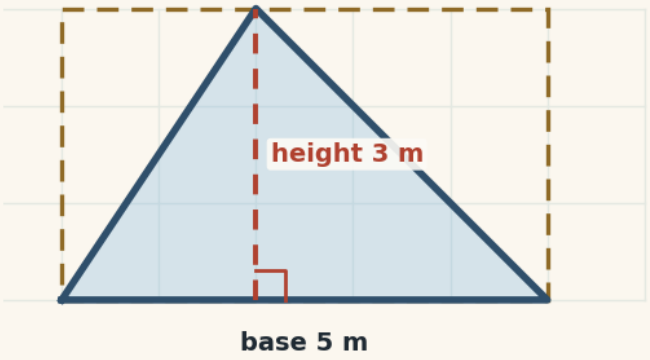


1. The rectangle round it: 8 × 4	32 m^2
2. Cut along the height; turn each piece into the empty corner beside it.	the pieces fill the rest exactly
3. So the triangle is half the rectangle.	½ × 32 = 16 m^2

Area = ½ × 8 × 4 = 16 m^2.

Example 2 — A half in the answer

Find the area of a triangle with base 5 m and height 3 m.



1. Base × height	5 × 3 = 15
2. Halve it.	15 ÷ 2 = 7½
3. Area	7½ m^2 (or 7.5 m^2)

Area = 7½ m^2.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (perimeter or area (Lesson 1), and multiplying), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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